

Name of Program	Bachelor of Technology (Computer Science Engineering)						
CSET333	Problem Solving using C				L	T	P
Owning School/Department	School of Computer Science Engineering and Technology				2	0	2
Pre-requisites/Exposure	-						

Course Outcomes (COs)

On completion of this course, the students will be able to:

CO1: To explain various concepts in C programming language and understand the problem-solving aspect.

CO2: To Implement concept of pointer and perform I/O operations in files and make use of concepts in finding solutions to real-life problems.

CO-PO/PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	2	2	1	2		2	1	1	2	1	2	1
CO2	2	2	2	1	2	1	1					2	2	3	1

1=weakly related

2= moderately related

3=strongly related

Course Contents:

Module I:

8 lecture hours

Program structure, Communications with the Operating System, Library Functions, Pre-processor Directives, Debugging and Efficiency, Memory Models, Data Types-Simple C data types, Integer Data types, Floating Point types, Derived Data types, The ASCII Character Set, Compilation and Linking, Types of compilers, Decision making, Loops, Switch statement, Goto statement, Null statement, Comma operator, setjmp functions, longjmp functions, Storage classes: automatic variable, external, static, register, Arrays, Array Indexing, Using Array names as Pointers, Character Arrays.

Modulo II:

10 lecture hours

Advanced data types: #define statement, Variable length array, Flexible array members, Complex number type, Type Qualifiers: Const, Volatile, Restrict, Functions, Pass by value, Pass by reference, Command Line Arguments, Structures, Arrays of Structures, Structures of Arrays, Structures of Structures, Bit Fields in Structures, offsetof(), Macro, unions, Typedef, Typecasting, Implicit type casting, Explicit typecasting, Pointers- Declaration, NULL pointers, Indirection, Pointer to array, Pointer to function, Pointer to structure, Arrays of Pointers, Passing pointers to functions.

Module III:

10 lecture hours

File handling, Recursive functions, Memory allocation in a recursive method, Dynamic memory allocation, Global Memory versus Local Memory, Error handling in C, Global Variable Errno, perror() and strerror(), Exit status, Divide by zero errors, Interfacing C with Python code: Calling C functions from Python, Calling python functions from C, Threads: Creating a thread, Passing arguments and

returning values, Common thread functions, Thread synchronization concepts, Mutexes, condition variables.

Module IV:

13 lecture hours

Generic Security Software's, Windows Firewalls, Linux Firewalls, Access Control Lists of Firewalls, Types and Examples of Firewalls, Network-based Honeypots and Trapdoors, Virtual Private Network and its Implementation, Network Intrusion Detection System, Network Intrusion Prevention System, Router Security, Switch Security, Proxy Server and its Configurations, Load Balancers, IPv6 and IPv6 Security, Secure Forwarding in Overlay Networks.

Studio Work / Laboratory Experiments:

Students will implement the following programs in C language on a Linux based platform using GCC compiler.

Text Books :

1. Gottfried Byron, *Outline of Programming with C* (4th ed.), McGraw Hill Education, 2018. ISBN 978- 0070145900.

Reference Books :

1. Balaguruswamy E., *Programming in ANSI* (8th ed.), McGraw Hill Education, 2018. ISBN 978-9351343202.

Assessment Scheme:

Components	Internal Assessment	Mid Term Exam	End Exam	Total
Weightage (%)	45	20	35	100